Evaluation Report

AI-Powered Telecom Network Operations Assistant
Test cases, retrieval results, DeepEval scores and sample outputs
(recommended deliverable 8)

Item	Value
Automated tests	222 pytest tests, all passing
Evaluation set	8 hand-made cases (tests/evaluation/eval_cases.json)
RAG metrics	DeepEval 4.x, judge: nvidia openai/gpt-oss-20b
Deterministic metrics	condition recall, retrieval precision, average precision, category, hypothesis, grounding
Raw results	docs/evaluation/eval_results_deepeval.json

Contents

Contents 2

1. Summary 3

2. Test cases 3

3. Retrieval results 4

4. DeepEval scores (judge: nvidia openai/gpt-oss-20b) 5

5. Machine-learning evaluation 6

6. Sample outputs 7

1. Summary

Metric	Score	Meaning
Condition recall	1.00	expected conditions recognised in the query
Retrieval precision	1.00	share of top-5 records showing all expected conditions
Average precision	1.00	relevant records ranked first
Category match	1.00	derived category as expected
Hypothesis in top 2	1.00	an expected hypothesis ranked 1st or 2nd
Grounded	0.75	evidence grounded and no hallucination flagged by the code checks and the LLM judge combined
Faithfulness	0.93	DeepEval, LLM-judged (6/8 cases scored)
Answer relevancy	0.89	DeepEval, LLM-judged (8/8 cases scored)
Contextual precision	1.00	DeepEval, LLM-judged (8/8 cases scored)
Contextual relevancy	0.28	DeepEval, LLM-judged (8/8 cases scored)

Read with care

These are **development scores**: one retrieval weight (metric similarity 0.6) was tuned on these eight cases. A held-out benchmark grown from reviewed feedback is future work.
The DeepEval judge is nvidia openai/gpt-oss-20b. A smaller judge model is stricter in places and can misread evidence; its scores are reported as returned, without adjustment.

2. Test cases

Eight cases, one per scenario in the project specification. Expected values follow the documented rules and are checked against the records actually retrieved — never against invented answers.

Case	Query	Expected conditions	Expected category	Expected hypotheses
latency_packet_loss	Cell has high latency and high packet loss. What similar network conditions occurred before?	high_latency, high_packet_loss	Service Degradation	transport_connectivity
signal_call_drops	Poor signal strength and frequent call drops on a cell.	weak_signal, high_call_drop	Radio / Signal Quality	radio_quality
load_throughput	High user load with reduced throughput on the cell.	high_load, low_throughput	Congestion / Load	capacity_congestion
low_handover	Handover success rate is low on several cells.	low_handover	Mobility / Handover	mobility_configuration
alarms	High alarm count and many critical alarms at a site.	high_alarm_count, high_critical_alarm_count	Fault / Alarm	equipment_fault, availability_outage
authentication	Many authentication failures reported on the network.	authentication_failure	Security	security_authentication
setup_success	Low connection setup success for users trying to connect.	low_setup_success	Access / Connection Setup	access_procedure
fault_rate	High fault occurrence rate in a network region.	high_fault_rate	Fault / Alarm	equipment_fault, weather_related

Automated test suite

- The five proofs the specification demanded are tests: no raw value fabricated (every processed value compared with the raw CSV text), filters never violated, retrieved IDs are real records, the LLM cannot add unsupported numbers (a deliberately hallucinating fake LLM), and no target leakage.
- Provider tests run without network access (fake clients), so the suite needs no API key.

Test file	Tests
`test_phase14_guardrails.py`	19
`test_phase15_18_rag.py`	36
`test_phase19_22_models.py`	31
`test_phase1_dataset.py`	7
`test_phase23_24_agents.py`	7
`test_phase25_evaluation.py`	9
`test_phase26_27_api.py`	19
`test_phase2_preprocessing.py`	31
`test_phase3_conditions.py`	8
`test_phase4_8_derived.py`	27
`test_phase9_13_retrieval.py`	28
Total	**222**

3. Retrieval results

Per case: the deterministic retrieval and diagnosis metrics, the derived category, the top hypotheses and the time for the full analysis including LLM calls.

Case	Cond. recall	Precision@5	Avg. precision	Category	Top hypotheses	Explanation	Time (s)
latency_packet_loss	1.00	1.00	1.00	Service Degradation	transport_connectivity, weather_related	llm	82
signal_call_drops	1.00	1.00	1.00	Radio / Signal Quality	radio_quality, weather_related	template	38
load_throughput	1.00	1.00	1.00	Congestion / Load	capacity_congestion, weather_related	llm	16
low_handover	1.00	1.00	1.00	Mobility / Handover	mobility_configuration, load_mobility	llm	10
alarms	1.00	1.00	1.00	Fault / Alarm	equipment_fault, availability_outage	template	46
authentication	1.00	1.00	1.00	Security	security_authentication, weather_related	template	26
setup_success	1.00	1.00	1.00	Access / Connection Setup	access_procedure, radio_quality	llm	20
fault_rate	1.00	1.00	1.00	Fault / Alarm	weather_related, equipment_fault	llm	15

Example retrieval (real): 'high latency and packet loss', filters Minnesota + Winter

Record	Hybrid	Semantic	Keyword	Metric	Conditions	State · season
ROW-033707	0.996	0.568	1.000	0.990	high_latency, high_packet_loss	MN · Winter
ROW-099453	0.959	0.534	0.973	0.985	high_latency, high_packet_loss, security_risk	MN · Winter
ROW-040582	0.944	0.523	0.972	0.972	high_latency, high_packet_loss, low_setup_success	MN · Winter

412 records matched the filters; 386 candidates were re-scored; weights used {'semantic': 0.375, 'keyword': 0.25, 'metric': 0.375}. Every returned record satisfies both filters.

4. DeepEval scores (judge: nvidia openai/gpt-oss-20b)

Case	Faithfulness	Answer relevancy	Contextual precision	Contextual relevancy
latency_packet_loss	1.00	0.59	1.00	0.42
signal_call_drops	1.00	1.00	1.00	0.28
load_throughput	0.60	0.92	1.00	0.37
low_handover	–	1.00	1.00	0.16
alarms	1.00	1.00	1.00	0.30
authentication	–	0.60	1.00	0.16
setup_success	1.00*	1.00	1.00	0.30
fault_rate	1.00*	1.00	1.00	0.24
Mean of scored cases	**0.93**	**0.89**	**1.00**	**0.28**

'–' means the judge could not return a valid answer after 3 attempts; such a metric is excluded from the mean instead of being counted as 0. '*' marks a score that needed a retry.

- **Faithfulness**: share of claims in the answer supported by the retrieved context.
- **Answer relevancy**: share of the answer's statements that address the question.
- **Contextual precision**: whether relevant retrieved records are ranked above irrelevant ones.
- **Contextual relevancy**: share of the retrieved context relevant to the question — lower by design, because each record's text lists every metric, most unrelated to a given question.

Comparison with the earlier run (judge: Claude Haiku 4.5, 9 Oct 2026)

Metric	Earlier run (Claude judge)	This run (nvidia openai/gpt-oss-20b)
Faithfulness	0.96	0.93
Answer relevancy	0.95	0.89
Contextual precision	1.00	1.00
Contextual relevancy	0.68	0.28

Both runs evaluate the same pipeline and retrieval (retrieval precision is identical); the LLM that writes the explanation and the judge both changed between them. The smaller open model writes less precise explanations and is a stricter, less consistent judge, so the faithfulness gap mainly reflects the models, not the evidence.

A first NVIDIA run (kept as eval_results_deepeval_nvidia_run1_unretried.json) recorded three judge failures — the judge's reply was not valid JSON — as scores of 0, which lowered the means. The runner now retries such replies and excludes metrics that still cannot be scored.

Judge explanations for the lowest scores

Case	Metric	Score	Judge's reason (abridged)
latency_packet_loss	AnswerRelevancy	0.59	The score is 0.59 because the response focuses on diagnosing and recommending actions for current congestion, corrective steps, and future reviews, rather than listing prior similar latency or packet loss events as requested.
load_throughput	Faithfulness	0.60	The score is 0.60 because the actual output incorrectly labels all five records as having low throughput, while the records actually show values between 268 and 298 Mbps, which are not low. Additionally, it claims the deterministic analysis confirms a congestion problem with low severity, yet the records classify the s
low_handover	Faithfulness	–	error: Evaluation LLM outputted an invalid JSON. Please use a better evaluation model.
authentication	Faithfulness	–	error: Evaluation LLM outputted an invalid JSON. Please use a better evaluation model.
authentication	AnswerRelevancy	0.60	The score is 0.60 because the response included several irrelevant statements about weather, connection setup, and resilience reviews that do not address the reported authentication failures, limiting the overall relevance.

Not every low score is a defect of the answer. In load_throughput the judge states that 268–298 Mbps is 'not low throughput', but the documented condition (thresholds.yaml) is below 304.248 Mbps, so the answer was right and the judge was wrong: it does not know the project's thresholds. In the application, the LLM judge is therefore combined with deterministic checks rather than trusted alone.

RAGAS

RAGAS measures the same dimensions (faithfulness, context precision/recall, answer relevancy). Running it with the same judge would duplicate this measurement, so it was not added; the guideline lists DeepEval or RAGAS.

5. Machine-learning evaluation

Model (served)	Test set (20%)	Result
Fault risk — HistGradientBoosting, feature set C	20,000 records	R^2 0.8849, MAE 3.7249, RMSE 4.7669; 5-fold CV R^2 0.8848 ± 0.0006
Throughput — HistGradientBoosting	20,000 records	R^2 0.3283, MAE 161.8 Mbps (mean baseline MAE 197.3)
Category classifier — HistGradientBoosting	20,000 records (stratified)	accuracy 0.9737, macro F1 0.9759 (agreement with the rule-derived labels)

Fault-risk feature set	Linear R^2	Gradient boosting R^2	Status
C_primary	0.8848	0.8849	served
D_strict	0.48	0.4783	strict reference
A_formula_demo	0.9998	0.9997	leakage demonstration, never served

6. Sample outputs

Real text report for the specification's example query (POST /api/v1/analyze, LLM off): *A network region has increasing fault occurrence and declining handover success. What corrective actions should be considered?*

```
INCIDENT CLASSIFICATION

Category: Mobility / Handover (secondary: Fault / Alarm)
Severity: MEDIUM (score 17.5, stated conditions (estimated))
Priority: P4 - Low (score 20.5)

FAULT RISK

Predicted risk: 44.08% fault occurrence (80% interval 38.17-50.01%; ML prediction, input coverage 0%)
Risk level: Medium
Note: No model inputs were supplied, so this is the model's baseline expectation, not a case-specific
    prediction. The historical rate of similar records is the stronger evidence here.
Historical fault occurrence of similar records (observed): mean 58.05%, range 56.25-61.21% over 5 records

SIMILAR HISTORICAL RECORDS

Record ID  | Similarity | Key Conditions | Category
ROW-098813 | 0.98 | low_handover, high_fault_rate | Mobility / Handover
ROW-058468 | 0.98 | low_handover, high_fault_rate | Mobility / Handover
ROW-069805 | 0.97 | low_handover, high_fault_rate | Mobility / Handover
ROW-047042 | 0.96 | low_handover, high_fault_rate | Mobility / Handover
ROW-051245 | 0.96 | low_handover, high_fault_rate | Mobility / Handover

EVIDENCE
• [USER_INPUT] High fault occurrence rate reported in the query
• [USER_INPUT] Low handover success reported in the query
• [HISTORICAL_EVIDENCE] Low handover success present in 5 of 5 similar records
• [HISTORICAL_EVIDENCE] High fault occurrence rate present in 5 of 5 similar records
• [RAW_DATA] ROW-098813: latency 120.16 ms, packet loss 4.16%, handover success 90.41%, fault rate 56.25%
• [RAW_DATA] ROW-058468: latency 116.47 ms, packet loss 4.86%, handover success 90.32%, fault rate 61.21%
• [RAW_DATA] ROW-069805: latency 105.21 ms, packet loss 6.13%, handover success 90.57%, fault rate 57.14%
• [DERIVED_FEATURE] Derived category Mobility / Handover, severity MEDIUM (stated conditions (estimated))
• [ML_PREDICTION] Predicted fault occurrence risk Medium (44.08%)
• [HISTORICAL_EVIDENCE] Historical fault occurrence of similar records: mean 58.05%

LIKELY ROOT CAUSES

1. Mobility configuration issue (neighbour relations or handover parameters) (hypothesis)
   Confidence: 0.65 (heuristic)
   Evidence: low_handover
   Historical: 100% of the 5 retrieved similar records show the key evidence
2. Weather-related service degradation (hypothesis)
   Confidence: 0.45 (heuristic)
   Evidence: no direct evidence in the query
   Historical: 100% of the 5 retrieved similar records show the key evidence
   Note: In this dataset weather is the strongest driver of throughput, latency, packet loss, call drops
       and fault occurrence (see historical evidence).
3. Load-related mobility degradation (hypothesis)
   Confidence: 0.3 (heuristic)
```

```
   Evidence: low_handover
   Historical: 0% of the 5 retrieved similar records show the key evidence
4. Equipment fault indicated by alarms (hypothesis)
   Confidence: 0.29 (heuristic)
   Evidence: high_fault_rate
   Historical: 0% of the 5 retrieved similar records show the key evidence
5. Radio coverage or signal-quality degradation (hypothesis)
   Confidence: 0.26 (heuristic)
   Evidence: low_handover
   Historical: 0% of the 5 retrieved similar records show the key evidence
6. Transport or inter-cell connectivity degradation (hypothesis)
   Confidence: 0.24 (heuristic)
   Evidence: low_handover
   Historical: 0% of the 5 retrieved similar records show the key evidence
   Note: The dataset contains no transport/backhaul counters, so this cannot be confirmed from the data.

WHAT TO CHECK

1. Handover attempts, successes and failures per source-target cell pair
2. Neighbour relation lists of the affected cells (missing or unwanted neighbours)
3. Mobility parameters of the affected cells compared with the network baseline
4. Recent parameter or configuration changes on the affected cells
5. Whether the degradation coincides with rain or snow periods
6. Weather exposure of the affected links and sites
7. User count and traffic volume on the source and target cells during the degradation
8. Whether target cells were near capacity when handovers failed
9. Active and historical alarms at the affected sites (type and frequency)
10. Fault occurrence trend for the affected cells
11. Signal strength and quality distribution in the affected area
12. Coverage holes or overshooting cells near the affected area
13. Transport / backhaul latency, loss and utilisation for the affected sites
14. Connectivity between neighbouring sites used for handover signalling

RECOMMENDED ACTIONS

1. Audit neighbour relationships and correct missing or incorrect neighbours
2. Compare mobility parameters with the network baseline and roll back deviating recent changes
3. Correlate the degradation with weather periods before changing configuration
4. Review weather resilience of exposed links and sites
5. Review load-balancing between neighbouring cells
6. Consider capacity relief (traffic offload or additional capacity) for persistently loaded cells
7. Prioritise resolution of critical alarms at the affected sites
8. Investigate recurring alarm types to find the failing component
9. Monitor high fault occurrence rate (fault_rate_pct) after remediation and compare it with the dataset
    baseline (median 39.46 over 100000 records)
10. Monitor low handover success (handover_success_pct) after remediation and compare it with the dataset
    baseline (median 95.02 over 100000 records)

ADDITIONAL EVIDENCE REQUIRED
…
```

Validation block of the same analysis:


```
{
 "evidence_grounded": true,
 "diagnosis_supported": true,
 "recommendation_relevant": false,
 "hallucination_detected": false,
 "score": 1.0,
 "method": "deterministic only",
 "issues": []
}
```

LLM-written answer evaluated by DeepEval (case latency_packet_loss, passed the grounding gate):

```
The cell is experiencing high latency and high packet loss, matching five recent Service Degradation
    records that all occurred during rain in summer or fall.  All five records show low availability
    (≈77 %) and fault rates around 44 %, with the dataset’s baseline indicating a 47.89 % fault rate for
    rain.  The most likely drivers are weather-related degradation and transport connectivity issues, while
    capacity congestion is less supported by the evidence.
High latency and packet loss are present in the operator’s stated conditions and in all retrieved records.
    The records share identical weather (Rain) and season (Summer/Fall) and exhibit similar metrics:
    latency ≈156 ms, packet loss ≈13 %, call drop ≈1.3 %, handover success ≈92 %, throughput ≈511 Mbps, and
    fault rate ≈44 %.  The dataset baseline shows a median latency of 77 ms and packet loss of 6 %,
    confirming the degradation.
Recommended actions: Investigate transport and inter-cell connectivity for the affected sites; Correct
    transport faults or capacity bottlenecks found during the investigation; Correlate the degradation with
    weather periods before changing configuration; Review weather resilience of exposed links and sites;
    Review capacity of the affected cells (load balancing, carrier/capacity expansion); Apply traffic
    management for persistently congested cells
```

The five README sample queries and their real outputs are in data/sample_data/sample_queries.json.